

Q₁ - Greedy algorithm for a Multi-Armed Bandit problem

Given

Machine A $\rightarrow [3, 3, 4]$

Machine B $\rightarrow [6, 7]$

Machine C $\rightarrow [5, 4, 6]$

Formula:

$$Q(a) = \frac{\text{Sum of rewards}}{N(a)}$$

Machine A:

$$Q(A) = \frac{3+2+4}{3} = 3.0$$

Machine B:

$$Q(B) = \frac{6+7}{2} = 6.5$$

Machine C:

$$Q(C) = \frac{5+4+6}{3} = 5.0$$

The Greedy option is B = 6.5.

Q2 - Greedy algorithm using incremental update method.

Given

machine A [3, 3, 4]

machine B [6, 7]

machine C [5, 4, 6]

Initially

$$Q(A) = 0$$

$$N(A) = 0$$

$$Q(B) = 0$$

$$N(B) = 0$$

$$Q(C) = 0$$

$$N(C) = 0$$

Formula

$$Q_{\text{new}}(a) = Q_{\text{old}}(a) + \frac{R - Q_{\text{old}}(a)}{N(a)}$$

Machine A :- Rewards :- 3, 3, 4

First reward = 3

initially:

$$Q(A) = 0$$

$$N(A) = 0$$

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$$Q(A) = 1$$

$$\text{update } Q(A) = 0 + \frac{3 - 0}{1}$$

$$Q(A) = 3$$

second R = 2

$$N(A) = 2$$

$$Q_{\text{old}}(A) = 3$$

$$Q(A) = 3 + \frac{2 - 3}{2}$$

$$Q(A) = 3 - \frac{1}{2}$$

$$Q(A) = 2.5$$

Third $R=4$

$$N(A) = 3$$

$$Q_{old}(A) = 2.5$$

$$Q(A) = 2.5 + \frac{4 - 2.5}{3}$$

$$Q(A) = 2.5 + \frac{1.5}{3}$$

$$Q(A) = 3.0$$

Final Machine

$$N(A) = 3$$

$$Q(A) = 3.0$$

$$Q_{new}(c) = 4.5$$

Third $R=6$

$$N(c) = 3$$

$$Q_{old}(c) = 4.5$$

$$Q(c) = 4.5 + \frac{6 - 4.5}{3}$$
$$= 4.5 + \frac{1.5}{3}$$

$$Q(c) = 5.0$$

Final Machine C

$$N(c) = 3$$

$$Q(c) = 5.0$$

Machine B (Rewards: -6, 7)

First $R=6$

$$N(B) = 1$$

$$Q_{old}(B) = 0$$

$$Q(B) = 0 + \frac{6 - 0}{1}$$

$$Q(B) = 6$$

Therefore

$$Q(A) = 3.0$$

$$Q(B) = 6.5$$

$$Q(c) = 5.0$$

Second $R=7$

$$N(B) = 2$$

$$Q_{old}(B) = 6$$

$$Q(B) = 6 + \frac{7 - 6}{2}$$

$$= 6 + \frac{1}{2}$$

$$Q(B) = 6.5$$

The highest value is

$$Q(B) = 6.5$$

Final Machine B

$$N(B) = 2$$

$$Q(B) = 6.5$$

Machine C (Rewards: -5, 4, 6)

First $R=5$

$$N(c) = 1$$

$$Q_{old}(c) = 0$$

$$Q_{new} = 5$$

Second $R=4$

$$N(c) = 2$$

$$Q_{old}(c) = 5$$